

# MUTHU D

AWS DevOps Engineer

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## PROFESSIONAL SUMMARY

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DevOps Engineer with 3 years of experience in automating cloud infrastructure, building CI/CD pipelines, and managing containerized application deployments on AWS. Proficient in Terraform, Jenkins, Docker, and Ansible, with hands-on knowledge of Kubernetes (EKS) for container orchestration and CD deployment activities. Experienced with ArgoCD, SonarQube, Nexus Repository, Prometheus, and Grafana. Committed to delivering reliable, scalable, and well-monitored infrastructure in agile DevOps environments.

## TECHNICAL SKILLS

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- **Cloud Platform:** AWS
- **CI/CD Tool:** Jenkins
- **GitOps Tool:** ArgoCD
- **Infrastructure as Code:** Terraform, Ansible
- **Containerization Tool:** Docker, Docker Compose, DockerHub
- **Orchestration Tool:** Kubernetes (EKS)
- **Build Tool:** Maven
- **Scanning / Code Quality Tool:** SonarQube
- **Artifact Repository:** Nexus Repository
- **Monitoring & Logging:** Prometheus, Grafana, AWS CloudWatch
- **Version Control & Collaboration:** Git, GitHub, GitLab, Jira
- **Scripting & Operating Systems:** Bash, Linux (Amazon Linux), Python (Basic)

## PROFESSIONAL EXPERIENCE

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**Lead Engineer – DevOps** | **HCL Technologies**

*Nov 2025 – Present*

*Bangalore, India*

- Built and managed CI/CD pipelines using Jenkins, integrated with GitHub for source code management and SonarQube for automated code quality checks, ensuring reliable and consistent build and test execution for applications deployed on AWS EC2.
- Provisioned and managed AWS infrastructure using Terraform modules and workspaces, including VPC, EC2, ALB, RDS, and S3, with remote state maintained in Amazon S3 to ensure consistent and version-controlled infrastructure across multiple environments.

- Built Docker images using multi-stage builds to reduce image size and improve build performance, pushed images to DockerHub and Nexus Repository, and managed image tagging with build numbers for traceability across the CI pipeline.
- Configured AWS networking components including VPCs, public and private subnets, route tables, NAT gateways, Internet gateways, and VPC peering connections to establish secure and well-structured network architecture for application environments.
- Set up monitoring and alerting using AWS CloudWatch, Prometheus, and Grafana, built dashboards to track system performance metrics, and configured alert rules for CPU, memory, and disk usage to proactively detect infrastructure issues.
- Used Ansible playbooks to patch and update EC2 instances, apply configuration changes, and maintain consistent server states across environments, reducing manual effort involved in routine infrastructure maintenance tasks.
- Supported senior engineers in CD deployment activities including application deployments to AWS EKS, assisting with deployment YAML updates, checking pod and node status, and tracking release tasks using Jira during deployment cycles.
- Performed Linux system administration tasks on AWS EC2 instances including user access management, package installations, and basic troubleshooting of service failures, ensuring smooth day-to-day operation of application servers.

## **Software Engineer** | **Xitadel CAE Technologies**

*May 2023 – Sep 2025*

*Bangalore, India*

- Set up and managed AWS EC2 instances, configured IAM roles and security groups with least-privilege access, and managed SSH key-based authentication for secure infrastructure access.
- Created and managed S3 buckets for file storage and archival, configured bucket policies and lifecycle rules to control data retention and ensure secure long-term storage.
- Configured AWS VPC networking including public and private subnets, route tables, internet gateways, and NAT gateways to build a secure and segmented network for cloud workloads.
- Set up AWS RDS instances for database hosting, configured automated backups, snapshot schedules, and parameter groups to ensure data availability and quick recovery from failures.
- Monitored AWS infrastructure using CloudWatch metrics and logs, created dashboards to track EC2 resource usage, and set up basic alarms to detect and respond to infrastructure issues.
- Containerized application toolchains using Docker and Docker Compose, enabling consistent and reproducible environments and eliminating dependency conflicts across the team.
- Managed version control using Git and GitHub, maintained branching strategies, reviewed code changes, and ensured clean commit history across project repositories.

- Wrote Bash scripts to automate routine Linux server tasks including log cleanup, disk monitoring, and scheduled backups, reducing manual effort for repetitive operations.
- Documented AWS infrastructure setup, configuration procedures, and operational steps to support team knowledge sharing and smooth onboarding of new members.

## KEY PROJECTS

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### CI/CD Pipeline – Terraform + Jenkins + Docker + AWS EKS

- Built CI/CD pipeline stages using Jenkins on EC2, building Docker images with multi-stage builds and pushing them to DockerHub and Nexus Repository as part of the automated build process.
- Provisioned AWS infrastructure including VPC, EC2, EKS cluster, and ALB using Terraform modules with S3 remote backend, maintaining consistent infrastructure across environments.
- Integrated SonarQube into the Jenkins pipeline for code quality checks; EKS deployment steps were handled by senior engineers using ArgoCD as part of the team's GitOps CD workflow.
- Managed Docker image versioning and tagging strategies, ensuring proper build number tagging for traceability and clean image management across DockerHub and Nexus repositories.

### GitOps Deployment using ArgoCD

- Supported the team's ArgoCD-based GitOps workflow where Kubernetes manifests in Git repositories were automatically synced to AWS EKS clusters on every approved code merge.
- Assisted with ArgoCD application health checks, tracked sync status across environments, and escalated sync failures or deployment drift issues to senior engineers for resolution.
- Gained practical understanding of GitOps principles — how Git-based state management provides audit trails, easy rollbacks, and consistent deployments across dev and prod clusters.

### Prometheus + Grafana Monitoring on AWS EC2

- Deployed Prometheus on AWS EC2 to scrape metrics from target instances, configured scrape intervals and retention settings, and connected Grafana as the visualization data source.
- Built Grafana dashboards covering CPU usage, memory consumption, disk I/O, and network traffic, giving the team a single-pane view of overall infrastructure health.
- Configured Prometheus Alertmanager with alert rules for critical thresholds and set up notification channels to reduce mean time to detection for infrastructure issues.

## EDUCATION

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### B.E. – Mechanical Engineering

Anna University, Tamil Nadu

2022